Visual Studio 2022 – Step by Step Guide

Create Solution, Web App, Web API, Class Library, and Push to Git Repository

1) Create a New Solution

Option A – Blank Solution:

1. Open Visual Studio 2022.
2. File → New → Project → search 'Blank Solution'.
3. Enter Solution name (e.g., DotnetTrainingDemo).
4. Choose a folder location.
5. Click Create.

Option B – Direct from Project Template:

1. File → New → Project → select any template (e.g., Web App).
2. Configure → Project name and location.
3. Uncheck 'Place solution and project in the same directory' for better structure.
4. Click Create.

2) Create a New Web Project

1. Right-click Solution → Add → New Project…
2. Choose 'ASP.NET Core Web App (MVC or Razor Pages)'.
3. Name it (e.g., WebApp).
4. Select Framework = .NET 8.0 (LTS).
5. Authentication = None, Configure for HTTPS = ✔.
6. Click Create.
7. Right-click project → Set as Startup Project.
8. Run with Ctrl+F5.

3) Create a New Web API Project

1. Right-click Solution → Add → New Project…
2. Select 'ASP.NET Core Web API'.
3. Project name = WebApi.
4. Framework = .NET 8.0.
5. Enable Controllers = ✔, Enable OpenAPI (Swagger) = ✔, Configure for HTTPS = ✔.
6. Click Create.
7. Set as Startup Project to run.
8. Run with Ctrl+F5 → Swagger opens at /swagger.

4) Create a New Class Library Project

1. Right-click Solution → Add → New Project…
2. Choose 'Class Library (.NET)'.
3. Project name = App.Core (example).
4. Framework = .NET 8.0.
5. Click Create.
6. Right-click WebApp → Dependencies → Add Project Reference… → tick App.Core → OK.
7. Use App.Core classes in WebApp code with 'using App.Core;'.

5) Push the Solution to Git Repository

Option A – Initialize & Publish from Visual Studio:

1. Open View → Git Changes.
2. Click 'Create Git Repository' if not already initialized.
3. Choose Local only or GitHub/Azure DevOps and sign in.
4. Select VisualStudio .gitignore preset, optionally add README.md.
5. Enter commit message (e.g., Initial solution + projects).
6. Click Commit All or Commit All and Push.
7. If only committed locally, click Push or Publish to remote.
8. Repository now visible on GitHub/Azure DevOps.

Option B – Clone remote first:

1. Create repo on GitHub/Azure DevOps first.
2. In VS: Git → Clone Repository → paste URL.
3. Add/Create solution inside cloned folder.
4. Commit and Push changes using Git Changes window.

Classroom Lab Exercise

Tasks for students:

1. Create a blank solution called LabDemo.
2. Add three projects: WebApp (MVC), WebApi, and a Class Library (Utilities).
3. Add a simple method in Utilities (e.g., Add two numbers).
4. Call the Utilities method inside WebApi and expose it via Swagger endpoint.
5. Commit and Push solution to GitHub.
6. Verify repository contents online.

Expected Output:

• Running WebApp shows MVC site homepage.
• Running WebApi shows Swagger page with sample endpoints.
• Utilities project code is used in WebApi.
• Repository on GitHub contains solution and projects with commits.